2.2.4 Transport systems in plants

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the need for transport systems in multicellular plants	To include references to size, variations in metabolic rate and the significance of surface area to volume ratio.
(b)	the structure, function and location of vascular tissue in roots, stems and leaves	To include xylem vessels, sieve tube elements and companion cells in the roots, stems and leaves of monocotyledonous crop plants (cereals) and dicotyledonous crop plants (broad-leaved crops e.g. carrots, potatoes).

(c)	**(i)**	the observation, drawing and annotation of stained sections of plant tissues using a light microscope	*M0.1, M0.2, M1.1, M1.2, M1.8, M2.1* **PAG1** HSW2, HSW3, HSW4, HSW5, HSW8
	(ii)	the longitudinal and transverse dissection and examination of plant organs to demonstrate the position and structure of vascular tissue	**PAG2** HSW2, HSW3, HSW4, HSW5, HSW8
(d)		the entry and transport of water in terrestrial plants	To include details of the pathways taken by water **AND** the mechanisms of movement, including adhesion, cohesion and the transpiration stream, in terms of water potential. HSW2, HSW8
(e)	**(i)**	the process of transpiration and the environmental factors that affect the transpiration rate	To include an appreciation that transpiration occurs due to physical processes linked to gaseous exchange in leaves.
	(ii)	practical investigations to estimate transpiration rates	*M0.1, M0.2, M1.1, M1.2, M1.3, M1.6, M1.11, M3.1, M3.2, M3.3, M3.5, M3.6, M4.1* **PAG5** HSW2, HSW3, HSW4, HSW5, HSW6, HSW8
(f)		the mechanism of translocation.	To include translocation in the phloem as an energy-requiring process transporting assimilates, especially sucrose, between sources (e.g. leaves) and harvestable sinks (e.g. roots, stems and seeds) **AND** details of active loading at the source and removal at the sink